WATER

📖 1. Definitions & Core Concepts Section

- **Water (H_2O):** The most important natural resource, acting as a major constituent of all living organisms (plants, animals, human beings) and the environment. 🌍
- **Aqueous Solutions:** Solutions formed when water dissolves other substances. 💧
- **Solvent:** A substance that acts as a medium in which other substances dissolve. 🧪
- **Universal Solvent:** A solvent (like water) capable of dissolving a remarkably wide range of substances, including solids, liquids, and gases. 🌌
- **Latent Heat of Fusion of Ice:** The amount of heat energy required to change a substance from its solid state to a liquid state exactly at its melting point, without any change in temperature. 🧊
- **Latent Heat of Vaporization of Water:** The energy required to change water into its vapour state at its boiling point, without any change in temperature. ☁️
- **Specific Heat Capacity:** The fixed amount of heat energy absorbed by 1 g of water when its temperature is heated through exactly 1°C. 🌡️
- **Solubility of a Gas:** Expressed as the volume of that gas dissolved in a certain specific volume of a solvent. 🫧

💡 2. The "Why" & Extra Key Facts Section

- **Why water is a "Universal Solvent":** Water molecules have a unique arrangement where the single oxygen atom carries a negative charge, and the two hydrogen atoms carry positive charges. This allows them to bind with many different molecules. Furthermore, water has a **high dielectric constant**, which reduces the electrostatic force of attraction between positive and negative ions, easily dissolving even inorganic (electrovalent) compounds. ⚡

- **Why water pollution happens:** The exact same characteristic that sustains life—water's ability to easily dissolve a wide range of substances—is the basic cause of water pollution. 🧵
- **Why water is used as a coolant (car radiators, desert coolers):** Because of its extremely high specific heat capacity, water can absorb massive amounts of heat energy with only a small rise in its own temperature. 🚗
- **Why coastal climates are moderate:** The high specific heat capacity of large water bodies modifies nearby land climates (warmer in winter, cooler in summer). Land and sea breezes happen due to this moderating property. 🏖️
- **Why lakes and rivers do not freeze suddenly:** This is entirely on account of water's incredibly high specific latent heat of solidification. 🌊
- **Why marine life survives in frozen ponds:** Thanks to water's anomalous expansion, when the surface freezes into ice (which is a bad conductor of heat), the water below the ice layer stays in a liquid state (reaching up to 4°C at the bottom), sustaining aquatic life. 🐟
- **Why ice floats:** 92 volumes of liquid water expand to become 100 volumes of solid ice upon freezing. This gives ice a relative density of 0.92, making it lighter than water. 🧊
- **Why steam burns are so severe:** 1 g of steam condensing causes much more serious burns than 100°C boiling water because the steam suddenly releases a massive amount of hidden energy (specific latent heat of condensation = 2268 J/g) directly onto the skin. 🔥
- **Why sweating cools our bodies:** The evaporation of sweat takes the required latent heat of vaporization directly from our body heat during hot weather. 😅
- **Why torrential rain is violent:** It is caused by the sudden, massive release of latent heat of condensation in the atmosphere. ⛈️
- **Why pressure cookers are used in hills:** On mountains, atmospheric pressure is low (e.g., water boils at just 70°C on Mount Everest). Food

doesn't cook properly at this low temperature. A pressure cooker increases internal pressure, artificially raising the boiling point. 🌰

- **Why ordinary tap water leaves concentric rings when boiled away:** Tap, river, and well water contain dissolved solids. When the water evaporates, these solids are left behind as visible rings. (Note: Rainwater and distilled water do not form rings because they lack these dissolved solids). 🚰
- **Why distilled water creates "etchings" on glass:** Though glass seems completely insoluble, if distilled water is kept in a sealed glass bottle for a very long time, extremely small traces of the glass actually dissolve into the water, etching the inner surface. 🔍
- **Why boiled or distilled water tastes flat/unpleasant:** Pure water is completely tasteless. The pleasant taste of regular drinking water comes entirely from dissolved matter like air, carbon dioxide, and essential minerals—which are easily expelled by boiling. 👅
- **Importance of Dissolved Salts/Minerals:**
    1. They are absolutely essential for the growth and development of plants. 🌱
    2. They add taste to drinking water. 💧
    3. They supply our bodies with essential minerals. 🏃

📊 3. Comprehensive Data & Charts Section

Property / Value	Exact Textbook Data
Chemical Formula & Name	H_2O ; Dihydrogen oxide
Molecular Mass Calculation	$1 \times 2 + 16 \times 1 = 18$ amu
Water Coverage on Earth	>70% of Earth's area

Property / Value	Exact Textbook Data
Fresh Water Supply	Hardly 2.5% of total water (includes polar ice caps/glaciers)
Human Body Water Content	Nearly 70% of our body weight
Maximum Density (at 4°C)	1 g/cm^3 or 1000 kg/m^3
Boiling Point at Normal Pressure	100°C at 760 mm Hg
Specific Latent Heat of Fusion (Ice)	336 J/g or 80 cal/g
Specific Latent Heat of Vaporization	2268 J/g or 540 cal/g
Specific Heat Capacity	4.2 J/(g°C) or 1 cal/(g°C)
Dissolved Oxygen (O_2) in Water	33% (compared to only 21% in ordinary air)
Dissolved Nitrogen (N_2) in Water	66% (Oxygen is more soluble in water than Nitrogen)
Dissolved Carbon Dioxide (CO_2)	1% in water
Aquatic Oxygen Usage Ratio	1 dm^3 (1 litre) of water contains nearly 40 cm^3 of dissolved O_2

Property / Value	Exact Textbook Data
Solubility Example: Hydrogen	0.02 volumes of H_2 per unit volume of water at 0°C
Solubility Example: HCl	506 volumes of HCl gas per unit volume of water at 0°C

📌 4. Characteristics & Conditions Section

- **Physical Nature:** Pure water is a clear, transparent liquid that is entirely colourless, odourless, and tasteless. 💧
- **States of Matter:** Water exists in all three physical states: solid (ice), liquid (water), and gas (water vapour). It occurs in both free and combined states in nature. ❄️ 💧 ☁️
- **Boiling Point (100°C at 1 atm):** The temperature when water's vapour pressure exactly equals atmospheric pressure.
    - **Increases** due to an increase in pressure OR the presence of dissolved impurities.
    - **Decreases** when atmospheric pressure is lowered.
- **Freezing Point (0°C at 1 atm):**
    - **Decreases** due to an increase in pressure.
    - **Decreases** due to the presence of dissolved impurities.
- **Anomalous Expansion:** A highly unusual physical property. When normal liquids cool, they contract. Water contracts normally until it hits 4°C. However, as it cools below 4°C down to 0°C, it disobeys normal rules and starts *expanding* instead. 🔄
- **Water Purification Steps:**
    1. **Filtration:** Removes suspended impurities.
    2. **Distillation:** Removes dissolved impurities.

3. **Chlorination:** Adding a calculated amount of chlorine removes bacterial impurities. 🦠

- **Laboratory Experiments Noted:**
    - *To prove tap water has dissolved solids:* Boil tap water on a watch glass over a beaker of boiling water; solid concentric rings remain after evaporation. 🧪
    - *To prove tap water has dissolved gases:* Heat water in a round-bottom flask; escaping gas bubbles can be collected in a graduated tube via downward displacement of water. 🫧

🚀 5. Reactions & Biological Processes Master List

- **Aquatic Plant Photosynthesis:**

Aquatic plants utilize the carbon dioxide dissolved in water to prepare their food in the presence of chlorophyll and sunlight.

$$6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$$

- **Shell Formation for Marine Organisms:**

Carbon dioxide dissolved in water reacts with limestone to initially form calcium bicarbonate.

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$$

Follow-up biological process: Marine organisms (like snails and oysters) then actively extract the calcium carbonate from this calcium bicarbonate to build their hard outer shells. 🐚

📖 1. Definitions & Core Concepts Section

- **Pure Substances:** Substances that are elements or compounds. 🧪
- **Mixture:** A combination of two or more substances (elements or compounds) which are physically mixed, not chemically combined with each other, and present in any proportion. 🥣
- **Homogeneous Mixture:** A mixture where the constituents are uniformly mixed and have a uniform composition throughout. 💧

- **Heterogeneous Mixture:** A mixture where the composition is not uniform throughout and has visible boundaries of separation between its constituents. 🪨
- **Solution:** A homogeneous mixture of two or more components whose composition may be gradually changed by changing the relative amounts of the components. ✨
- **Solvent:** The medium of dissolution that allows one or more components to dissolve in it to form a solution. 🌊
- **Solute:** The substance that dissolves in the solvent to form a solution. 🧂
- **Binary Solution:** A solution made up of exactly two components (one solute and one solvent). ✌️
- **Ternary & Quaternary Solutions:** Solutions made up of three or four components, respectively. 🔢
- **Aqueous Solution:** A solution produced by dissolving a substance specifically in water. 🚰
- **Non-aqueous Solution:** A solution made in liquid solvents other than water (like alcohol, petrol, ether, benzene, carbon disulphide, or liquid ammonia). ⛽
- **True Solution:** A solution in which the size of the solute particles is extremely small, i.e., less than 10^{-9} m (1 nanometer). 🔬
- **Suspension:** A heterogeneous mixture in which solid particles are spread throughout the liquid without dissolving in it, and they settle down if the solution is left undisturbed. ⏳
- **Colloidal Solution (or Sols):** A heterogeneous mixture in which microscopically insoluble dispersed particles (10^{-9} m to 10^{-6} m in size) are suspended throughout another substance called the dispersion medium. 🌫️
- **Dispersed Phase:** The component present in a smaller amount in a colloidal solution. 👊
- **Dispersion Medium:** The component present in a larger amount in a colloidal solution. 👐

- **Tyndall Effect:** The phenomenon where a colloidal solution scatters a beam of light passing through it, making the path of light visible. 🔦
- **Brownian Motion:** The continuous zig-zag path of particles in a colloidal solution, visible under a microscope (observed by Robert Brown in 1828). 〰️
- **Emulsions:** Colloidal solutions in which the dispersed phase and the dispersion medium both are liquids. 🥛

💡 2. The "Why" & Extra Key Facts Section

- **How to identify the solvent if phases differ:** If components exist in different physical states, the solvent is the component that exists in the same phase (state) as the final solution. For example, sugar dissolved in water makes a liquid solution, so liquid water is the solvent. 🧊➡️💧
- **How to identify the solvent if phases are the same:** If two substances are in the same phase, the component that is in excess is termed the solvent. 📈
- **Why a microscope is needed for some mixtures:** In a homogeneous mixture, constituent particles cannot be seen without a powerful microscope, whereas in heterogeneous mixtures, boundaries can be seen with the naked eye. 👁️
- **Why "dilute" and "concentrated" terms are limited:** They do not provide any actual quantitative information. A dilute solution becomes concentrated just by adding more solute, and a concentrated one becomes dilute by adding more solvent. 🧪
- **Why lime water turns milky with CO_2:** The milky appearance is because solid particles of calcium carbonate are formed. These particles have a size of more than 1000 nm, forming a suspension that will slowly settle at the bottom if transferred to a beaker. 🥛
- **Why colloidal particles do not settle down:** Colloidal particles carry an electric charge. Because they have similar charges, they constantly repel one another and remain dispersed instead of settling like suspension particles. ⚡

- **How to separate colloidal particles:** They easily pass through ordinary filter paper, but they cannot pass through ultrafilters such as semipermeable animal membranes. A special method called **centrifugation** can be used to separate them. 🌪️
- **Why we see sunbeams in forests or dusty rooms (Tyndall Effect):** The particles of dust/smoke in a room, or mist/water droplets acting as colloidal particles in a dense forest canopy, are big enough to scatter the light, making the light beam's path visible. 🌲🌞
- **How emulsions mix:** In emulsions, the liquid constituents are generally not miscible (like oily/oil-soluble and water-soluble layers). They form separate layers and do not mix without the addition of certain substances known as **emulsifiers**. 🧴
- **Solid Solutions (Alloys):** A homogeneous solution of a solid into another solid. Gold used in jewelry is a solid solution containing gold and small quantities of copper or silver. Brass contains 70% copper (solvent) and 30% zinc (solute). 🥇

📊 3. Data & Charts Section

Mixture Particle Sizes:

Mixture Type	Particle Size Range
True Solution	Less than 1 nm ($< 10^{-9}$ m)
Colloidal Solution	Between 1 to 1000 nm (10^{-9} to 10^{-6} m)
Suspension	*Greater than 1000 nm ($> 10^{-6}$ m) (Note: One list item states >100 nm, but diagrams and main text specify >1000 nm)*

Common Solutes and Non-Aqueous Solvents:

Solute	Solvent
Chlorophyll	Methylated spirit
Iodine	Alcohol *(Forms Tincture of Iodine)*
Paraffin wax	Turpentine oil
Sulphur	Carbon disulphide
Pigments in paint	Turpentine oil
Phosphorus	Carbon disulphide
Pigments in nail polish	Acetone
Oils in grease	Petrol
Bromine	Carbon tetrachloride

Types of Colloidal Solutions (Table 3.1):

Name of Colloidal Sol	Dispersed Phase	Dispersion Medium	Examples
(i) Emulsion	Liquid	Liquid	Cod liver oil, tonics in liquid form, milk
(ii) Gel	Liquid	Solid	Butter, cheese, jelly, curd

Name of Colloidal Sol	Dispersed Phase	Dispersion Medium	Examples
(iii) Liquid aerosol	Liquid	Gas	Cloud, fog, mist
(iv) Foam	Gas	Liquid	Soda water, soap water
(v) Solid foam	Gas	Solid	Rubber, bread, pumice stone
(vi) Solid sol	Solid	Solid	Ruby glass, alloys
(vii) Sol	Solid	Liquid	Paints, gold sol, starch in water
(viii) Solid aerosol	Solid	Gas	Dust storm, smoke, fumes, vehicular exhaust

📌 **4. Characteristics & Conditions Section**

- **Characteristics of Homogeneous Mixtures:**
  - Constituents are uniformly mixed with uniform composition.
  - Examples: Air (nitrogen, oxygen, carbon dioxide, water vapour, inert gases), Sugar solution.
- **Characteristics of Heterogeneous Mixtures:**
  - Composition is not uniform and has visible boundaries.
  - Examples: Oil and water (separate layers), Sodium chloride and sand (clear boundaries), Sand mixed with water (sand settles).
- **Ways to Form a Solution:**
  - Liquid + Liquid = Water and alcohol.

- Solid + Liquid = Sugar solution, Iodine in alcohol.
- Liquid + Gas = Soda water.
- Solid + Solid = Alloys.

- **Characteristics of a True Solution:**
  1. It is a homogeneous mixture.
  2. The solute particles are very small, less than 10^{-9} m.
  3. It is clear and transparent.
  4. It does not scatter light.
  5. Its components cannot be separated by filtration.
  6. The solute particles in a solution do not settle down.
  7. Due to the very small size, the solute particles do not scatter a beam of light made to pass through it.
- **Characteristics of a Suspension:**
  1. It is a heterogeneous mixture.
  2. Solute particles are large, more than 100 nm (or >1000 nm in main text).
  3. Solute particles can be seen with the naked eye.
  4. The solid particles can be easily separated by ordinary filter paper.
  5. The particles settle down after some time under the influence of gravity.
     - Examples: Muddy water, chalk-water mixture, some paints, and medicines.
- **Characteristics of Colloidal Solutions:**
  1. It is a heterogeneous mixture, though it appears homogeneous.
  2. The size of solute particles ranges from 1 to 1000 nm, which is intermediate between true solutions and suspensions. Particles can be seen only under a microscope.

3. It is a two-phase system consisting of a dispersed phase and a dispersion medium.
4. Particles pass through ordinary filter paper like true solutions but cannot pass through ultrafilters like animal membranes.
5. Particles carry an electric charge and repel one another, remaining dispersed.
6. Particles move in a continuous zig-zag path (Brownian motion).
7. Colloids scatter a beam of light passing through them (Tyndall effect).

🌟 What is the Tyndall Effect?

Imagine shining a flashlight or a laser pointer through a glass of pure water. 💧 You wouldn't actually see the beam of light traveling *inside* the water; you would only see a bright dot on the wall behind the glass. 🧱 But, if you shine that exact same light through a glass of water with a few drops of milk mixed in, the whole beam suddenly lights up and glows! 🥛✨ That visible, glowing pathway of light? That is the **Tyndall Effect** in action! 🔦

🔬 The "Chemical Flow" (Why it happens)

To understand the flow, we have to look at the size of the particles floating in the mixture. 🔍

- **In a True Solution (like saltwater):** The dissolved salt particles are incredibly tiny (less than 1 nanometer). 🧂 They are so ridiculously small that light waves just zoom right past them without hitting anything. 🏎️💨 Because the light doesn't hit anything, it remains invisible to your eyes. 🙈
- **In a Colloidal Solution (like milk, mist, or dusty air):** The floating particles are just a little bit bigger (between 1 and 1000 nanometers). 🌫️ They aren't heavy enough to settle down at the bottom of the glass, but they are just big enough to act like millions of tiny floating mirrors. 🪞
- **The "Flow" of Light:** When a beam of light flows into a colloid, the light waves physically crash into these floating particles. 💥 When they crash, the light bounces and scatters in every single direction. 🎆 This

continuous scattering of light by the particles is what makes the actual path of the beam visible to your eyes! 👀

🎥 Scene 1: The Invisible Laser

Imagine you are standing in a pitch-black, completely empty room. 🌑 There is zero dust in the air. It is perfectly clean.

You turn on a laser pointer and aim it at the opposite wall. 🔴

Light naturally travels in a perfectly straight line. 📏 Because there is *absolutely nothing* floating in the air for the light to hit, you will NOT see the beam in the air. 🙈 You will only see a red dot on the far wall. Light is completely invisible from the side unless it bounces into your eye! 👁️

🎥 Scene 2: The Microscopic Disco Balls (The Particles)

Now, imagine someone walks in and shakes a really dusty rug. 🧹 Suddenly, thousands of tiny dust particles are floating in the air. 🌫️

Think of every single one of those floating dust particles as a microscopic **disco ball**. 🪩

🎥 Scene 3: The Crash & The Bounce

You turn the laser pointer back on. 🔴

The straight line of light shoots forward, but this time, it *crashes* right into those floating disco balls (the dust). 💥

When the light hits a dust particle, it doesn't just stop. It *bounces off* the surface of the dust particle in random directions—up, down, left, right, and importantly, straight into your eyes! 🔀

This bouncing in all directions is what scientists call **scattering**. 🎆 The particle caught the light, reflected it off its surface, and threw it everywhere.

🎥 Scene 4: The "Glowing Pillar"

So, why does it look like a solid "pillar" or a "beam" of light? 🤔

The only dust particles that are bouncing light are the ones *directly inside the path* of the laser. 🚦

The dust in the rest of the room stays completely dark. 🌑

So, what you are actually seeing isn't the light itself. You are seeing millions of tiny, illuminated dust particles lined up back-to-back exactly where the light is traveling. ✨

Because they are all clustered together in the shape of the laser beam, it looks to your brain like a solid, glowing 3D tube or pillar made of light! 🏛️

Let's Summarize the "How":

1. Light travels straight. ➡️
2. Light physically hits a floating particle in its path. 💥
3. The particle acts like a tiny mirror, bouncing the light sideways into your eye. 🪞👁️
4. You see that specific particle light up. 🌟
5. Millions of these lit-up particles clustered together look like a visible, glowing beam of light! 🔦

Imagine trying to mix liquid oil and liquid water. 💧🌻 No matter how much you shake them, they naturally hate each other and separate into two layers.

An **emulsion** is what happens when you completely force them to get along! 🤝

It is a special type of mixture where you break one liquid (like the oil) into millions of microscopic droplets and perfectly trap them floating suspended inside another liquid (like the water). 🪄✨

Because *both* the floating particles and the background medium are **liquids**, it looks completely smooth to your naked eye. 👀

The real-world feel: Think of **milk**! 🥛 To your eye, it is just one smooth white liquid. But microscopically, it is actually millions of tiny liquid fat drops permanently suspended inside liquid water. 🧈💧

*(Note: Because two liquids like oil and water naturally want to separate, they usually need a chemical "peacemaker" called an **emulsifier** to hold them together permanently!)* 🧼

📖 1. Definitions & Core Concepts Section

- **Coagulation:** The process of neutralization of charge on colloidal particles by the addition of some oppositely charged substance. ⚡
- **Saturated Solution:** A solution that cannot dissolve any more of the solute at a given temperature. 🛑
- **Unsaturated Solution:** A solution in which more of the solute can be dissolved at a given temperature. ➕
- **Supersaturated Solution:** A solution that holds more solute than it can theoretically do at a given temperature. 🍯
- **Solubility:** The amount of solute that dissolves in 100 g of solvent to form a saturated solution at a particular temperature. Formula: $Solubility = \frac{Mass\ of\ solute}{Mass\ of\ solvent} \times 100$. 🧮
- **Insoluble Substance:** A substance that has negligible solubility (e.g., silver chloride with 0.000015 g solubility). ❌
- **Sparingly Soluble:** A substance that has more than negligible but less than high solubility (e.g., calcium hydroxide with 0.17 g solubility). 👌
- **Soluble Substance:** A substance having high solubility (e.g., sodium chloride). ✅
- **Solubility Curve:** A line graph that plots changes in solubility of a solute in a solvent against changing temperature. 📈

💡 2. The "Why" & Extra Key Facts Section

- **Why alum stops bleeding:** Blood consists of haemoglobin, which are charged colloidal particles. Alum dissolves to provide ions that neutralize the charge on these particles (coagulation), causing the blood to become thick and stop flowing. 🩸
- **Why soap cleans dirty clothes:** Soap forms an emulsion between water and oily dirt drops, effectively removing dirt that ordinary water cannot. 🧼

- **Why the sky is blue:** Colloidal particles in the atmosphere (air molecules, dust, water vapour) scatter shorter wavelength blue light more strongly due to the Tyndall effect. 🌤️
- **Why deltas form:** River water carries suspended colloidal particles (clay, sand) that collide with sea water, causing coagulation and settling to form new land. 🏝️
- **Why you can cool a saturated solution slightly without crystals forming:** It can enter a metastable supersaturated state if no suspended impurities (like dust) are present and the solution is left completely physically undisturbed. 🤫
- **Why endothermic/exothermic processes matter:**
    - In an **endothermic process** (like dissolving KNO_3), solubility increases with an increase in temperature. 🔥
    - In an **exothermic process** (like dissolving calcium sulphate), solubility actually decreases as temperature increases (it prefers lower temperatures). ❄️

📊 3. Data & Charts Section

Substance	Solubility Example Data
Potassium Nitrate (KNO_3)	31.6 grams at 20°C; 108 grams at 60°C.
Glauber's Salt (Anomalous Peak)	Reaches peak solubility precisely at 32.8°C before falling.

📌 4. Characteristics & Conditions Section

- **Factors Affecting Solubility Rate:**
    1. **Size of solute particles:** Smaller size means greater surface area, which leads to a greater solubility rate.
    2. **Stirring:** Brings more solvent in contact with the solute, increasing the rate of formation.

3. **Temperature:** The solubility of most solids increases with a rise in temperature, whereas gas solubility in a liquid always decreases with a rise in temperature.

- **Effects on a Saturated Solution:**
    - **Effect of Cooling:** Solubility usually decreases; the dissolved solute separates out in the form of crystals, leaving behind a solution that is still saturated at that lower temperature.
    - **Effect of Heating:** The solution becomes unsaturated, meaning more solute can now be dissolved in it.
- **Uses of Solubility Curves:**
    1. To determine the solubility of a substance at a particular temperature.
    2. To find the effect of cooling hot solutions (to see how much salt will crystallize out).

(Note: There were no chemical equations on these pages, so the Reactions section is skipped to keep your notes clean!) 🚫🧪

📉 Understanding the Solubility Curves (The "Why")

Here is the exact explanation for the three compounds in your syllabus and the anomalous curve, breaking down why their lines on the graph look the way they do! 🧠🔍

1. Potassium Nitrate (KNO_3) & Sodium Nitrate ($NaNO_3$)

- **What the curve does:** It shoots upward steeply as the graph moves right.
- **Why it happens:** These substances show a considerable increase in solubility with a rise in temperature. This is because dissolving them is an **endothermic process**. They *absorb* heat to dissolve, so giving them more heat (higher temperature) makes them dissolve much better! 📈🔥

2. Sodium Chloride ($NaCl$)

- **What the curve does:** It is a nearly flat, straight horizontal line across the graph.

- **Why it happens:** The solubility of sodium chloride increases only a very little bit with an increase in temperature. Heat just doesn't affect it much compared to other salts. 🧂➡️

3. **Calcium Sulphate ($CaSO_4$)**
   - **What the curve does:** It goes *downward* as temperature increases.
   - **Why it happens:** This is the opposite of KNO_3. The solubility of calcium sulphate decreases as temperature increases because dissolving it is an **exothermic process**. It releases heat when it dissolves, so adding more external heat actually slows the process down. *(Note: Hydrated calcium sulphate decreases after hitting a certain temp, while anhydrous decreases immediately with any temp rise).* 📉❄️
4. **Anomalous Solubility (Glauber's Salt: $Na_2SO_4 \cdot 10H_2O$)**
   - **What the curve does:** It goes up like a mountain, peaks exactly at 32.8°C, and then falls down on the other side.
   - **Why it happens (The Anomaly):** Some salts have solubilities that first increase and then decrease. For Glauber's salt, it rises steadily up to 32.8°C because it is **hydrous** (contains water molecules attached to it) below that temperature. However, once it crosses 32.8°C, it loses that water and becomes **anhydrous**, changing its chemical nature, which makes its solubility fall slightly at higher temperatures. 🌰💧

Let's totally pause the notes and just get the *feel* of this! 🛑✋ Coagulation sounds like a heavy science word, but it is actually a super simple and really cool concept. Let's break it down like a movie scene! 🎬🍿

🧲 The Setup: Why don't the particles just sink?

Imagine a swimming pool filled with millions of tiny floating particles (like dust or clay). Why don't they just sink to the bottom? 🏊

Because in a colloid, all these tiny particles carry the *exact same electrical charge* (let's say they are all negative).

Do you know how if you push two identical magnet poles together, they repel and push each other away? 🧲 That is exactly what is happening in the liquid! The

particles are constantly pushing away from each other, bouncing around, and staying afloat. They refuse to clump together. 🤷‍♂️

💥 Coagulation: The Crash and Sink!

Now, what happens if we sprinkle in a chemical that has the *opposite* charge (positive)? ⚡

Those new positive charges grab onto the negative particles and **neutralize** them. Suddenly, the particles lose their "magnetic forcefield." They stop repelling each other! 😳

Because they aren't pushing each other away anymore, they crash into each other, stick together into big, heavy blobs, and sink straight to the bottom. 📉

That clumping together and sinking is **coagulation**! 🎯

🩸 Explaining Your Book's First Example: Alum and Blood

Have you ever seen someone put a block of alum (*phitkari*) on a small shaving cut to stop the bleeding? Here is the invisible magic happening: ✨

1. **The Flowing Blood:** Blood is actually a colloid! The haemoglobin (red blood cells) are those tiny charged particles floating around. Because they all have the same charge, they keep flowing out of the cut. 💉
2. **The Magic Alum:** When you apply alum to the wet blood, it dissolves and releases ions with the *opposite* electrical charge. 🪄
3. **The Coagulation:** Those opposite charges immediately neutralize the blood particles. The blood cells stop repelling each other, violently clump together into a heavy traffic jam, and become thick. 🛑 This thick clump is a blood clot, and boom—the bleeding stops! 🩹

🏝️ Explaining Your Book's Second Example: The River Delta

This one is happening on a massive, geographical scale! 🌍

1. **The Muddy River:** As a river flows, it carries tons of tiny, floating clay and sand particles. They stay floating because they are charged colloids repelling each other. 🌊

2. **Meeting the Sea:** The river eventually flows into the ocean. What is ocean water full of? Salt! 🧂 Salt is packed with strong, oppositely charged ions.
3. **The Coagulation:** The moment the muddy river touches the salty sea, the salt neutralizes the river's clay particles. The clay stops repelling, clumps together into heavy mud, and sinks straight to the ocean floor right where the river ends. ⚓
4. **The Result:** Over hundreds of years, this massive pile of sunken, coagulated mud builds up and actually pokes out of the water, forming a brand new piece of land called a **Delta**! 🏝️

I hear you loud and clear! 🛑 Absolutely zero skipped points, zero skipped tables, and zero shortcuts. We are on a tight 4-hour clock, so I am giving you the **100% complete, unfiltered, exhaustive notes** straight from these exact pages. Let's lock in! 💯🔥

📖 1. Definitions & Core Concepts Section

- **Henry's Law:** At any given temperature, the mass of a gas dissolved by a fixed volume of a liquid is directly proportional to the pressure on the surface of the liquid.
- **Crystal:** A homogeneous solid of definite geometrical shape. It has symmetrically arranged smooth plane surfaces which meet forming the sharp edges.
- **Crystallisation:** A process by which crystals of a substance are obtained by cooling a hot saturated solution.
- **Water of Crystallisation:** The fixed amount of water that is associated with some compounds which is an integral part of their crystalline structure is called water of crystallisation.
- **Seeding:** This process of inducing crystallisation by adding a crystal of a pure substance into its saturated solution is called seeding.
- **Hydrated Substances:** Substances which contain water molecules along with salts like sodium carbonate decahydrate ($Na_2CO_3 \cdot 10H_2O$) and copper sulphate pentahydrate ($CuSO_4 \cdot 5H_2O$) are hydrated substances.

- **Anhydrous Substances:** A substance is anhydrous if it contains no water, for example, salts lacking their water of crystallisation. The term 'Anhydrous', is most often applied to crystalline substances when the water of crystallisation is removed. It can also refer to the gaseous form of some concentrated solutions such as ammonia.

💡 2. The "Why" & Extra Key Facts Section

- **Why soda water bubbles:** When the water surface is subjected to higher pressure, a lot more of CO_2 gas dissolves in it. On opening the soda water bottle, the dissolved gas rapidly bubbles out since pressure on the surface of the water suddenly decreases.
- **Why boiling makes water tasteless:** An increase in temperature of water causes a decrease in solubility of a gas in it. Since some taste of water is due to the gases in it, on boiling, these gases escape from water, leaving it tasteless.
- **Why chilled soft drinks give out less fizz:** A decrease in temperature of water increases the solubility of gases in water, that is why chilled soft drink bottles gives out less fizz.
- **Why crystals have beautiful colours:** In some cases, their beautiful colours are due to traces of water or impurities present in them. For example, the red colour of ruby is due to a trace of mercury oxide present in its crystals.
- **Important Caution for Crystals:** Avoid exposure to dust particles, crystals will form around them as well.

📊 3. Data & Tables Section

Table 3.3 : Solubility of solids and gases in water

	Solids	**Gases**
Temperature	The solubility of solids generally increases with temperature.	The solubility of gases decreases with temperature.

	Solids	Gases
Pressure	The solubility of solids does not depend on pressure.	The solubility of gases increases with pressure.

Table 3.4 : Some salts containing water of crystallisation

Common name	Chemical name	Formula	Colour
Washing soda crystals	Sodium carbonate decahydrate	$Na_2CO_3 \cdot 10H_2O$	White
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4 \cdot 7H_2O$	White
Potash alum	Hydrated potassium aluminium sulphate	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$	White
Glauber's salt	Sodium sulphate decahydrate	$Na_2SO_4 \cdot 10H_2O$	White
Blue vitriol	Copper (II) sulphate pentahydrate	$CuSO_4 \cdot 5H_2O$	Blue
Green vitriol	Iron (II) sulphate heptahydrate	$FeSO_4 \cdot 7H_2O$	Bluish green
Plaster of Paris	Calcium sulphate hemihydrate	$CaSO_4 \cdot \frac{1}{2}H_2O$	White
White vitriol	Zinc sulphate heptahydrate	$ZnSO_4 \cdot 7H_2O$	White

Common name	Chemical name	Formula	Colour
Gypsum	Hydrated calcium sulphate	$CaSO_4 \cdot 2H_2O$	Colourless

📌 **4. Characteristics & Exhaustive Lists Section**

- **Shapes in which crystals exist:**
  - Cubic (e.g., $NaCl$)
  - Rhombohedric (e.g., $CuSO_4$)
  - Octahedral (e.g., $FeSO_4$)
  - Prismatic, with sharp edges (e.g., KNO_3)
- **Methods to obtain crystals in a laboratory:**
  1. By cooling a hot saturated solution
  2. By slowly evaporating a saturated solution
  3. By cooling a fused mass
  4. By sublimation
- **Examples of Anhydrous Substances:**
  - Table salt is anhydrous sodium chloride.
  - Gaseous HCl is anhydrous.
  - Substances which do not contain any water along with the salts are anhydrous substances. For example sodium chloride ($NaCl$), potassium nitrate (KNO_3), etc.
- **Methods to obtain anhydrous salts from hydrated salts:**
  - Direct heating.
  - Heating in dry and hot air.
  - Heating under vacuum.

- By using dehydrating/desiccating agents such as warm concentrated sulphuric acid.

- **Mathematical Formula for Determination of water of crystallisation:**
  - $\%\ of\ water\ of\ crystallization = \frac{a-b}{a} \times 100$ *(where 'a' is weight of crystals at room temperature and 'b' is weight of residue after heating at 101°C).*

🧪 5. Reactions Master List Section

- **Heating Hydrated Copper Sulphate (Blue Vitriol):**

$$CuSO_4 \cdot 5H_2O \rightleftharpoons CuSO_4 + 5H_2O$$

(Hydrated blue colour salt heated turns to Anhydrous white colour salt powder. Cooling it reverses it).

You got it! 💯 I am giving you the absolute 100% complete, exhaustive notes from these specific pages, without skipping a single example, reaction, or table cell. You are going to crush this study session! 🚀📚

📖 1. Definitions & Core Concepts Section

- **Decrepitation:** On heating, crystals of potassium chloride produce a crackling sound, called decrepitation. This is due to the breaking of bigger crystals of potassium chloride into smaller ones.
- **Efflorescence:** A phenomenon where a compound loses its water of crystallisation on exposure to dry air is called efflorescence. This results in loss of crystalline shape, and finally the crumbling of crystals into a powder.
- **Deliquescence:** Certain water soluble substances, when exposed to the atmosphere at ordinary temperatures, absorb moisture from atmospheric air, become moist, lose their crystalline form, and ultimately dissolve in the absorbed water, forming a saturated solution.
- **Hygroscopy:** Certain substances absorb moisture (water vapour) from the atmosphere when they are exposed to it but not enough to form solutions.

- **Drying agents (Desiccants):** Drying agents are substances that can readily absorb moisture from other substances without chemically reacting with them.
- **Dehydrating agents:** Dehydrating agents are substances that can remove even the chemically combined water molecules from compounds.
- **Desiccator:** A desiccator is an air-tight glass vessel with a suitable drying agent (anhydrous calcium chloride) placed at the bottom. It is used for drying the solids placed in it.

💡 **2. The "Why" & Extra Key Facts Section**

- **Why efflorescence increases with heat:** The higher the temperature of the air, the higher the efflorescence. This is because air absorbs more water with rising temperature and decreasing moisture.
- **Why deliquescence is minimized in dry conditions:** Deliquescence occurs when vapour pressure inside the crystals is very low compared to vapour pressure in the atmospheric air. Thus, deliquescence is minimized during dry conditions. **Understanding "Vapor Pressure" in Deliquescence** 🧠💧
- Let's break this down into super easy language so you can actually *feel* what is happening!
- Imagine "Vapor Pressure" simply as the **pushing power of water moisture**. 🌊💪
- **The Atmospheric Air (Outside):** Think about a humid, rainy day. The air is absolutely packed with invisible water droplets. Because there is so much water in the air, its pushing power (Vapor Pressure) is **VERY HIGH**. ☁️🌧️
- **The Crystal (Inside):** Deliquescent crystals are extremely "thirsty." They have almost no water inside them pushing out. So, the pushing power (Vapor Pressure) inside the crystal is **VERY LOW**. 🧱🧊
- **The Science Rule:** Nature always moves things from a place of **HIGH pressure to LOW pressure**. 💨➡️
- **What happens? (The Deliquescence Effect)** 🫠

- Because the air is pushing moisture with high pressure, and the crystal has low pressure (no resistance), the water from the air gets forced *into* the crystal. The crystal acts like a dry sponge, aggressively sucking up the moisture from the air until it literally drowns in it and melts into a liquid puddle! That melting process is deliquescence. 🧽💦
- **Why is it minimized during dry conditions?** 🌞🌵
- If the weather is extremely dry, there is barely any moisture in the air. This means the atmospheric vapor pressure is also very low. Since the air doesn't have that strong pushing power to force water into the crystal, the crystal cannot absorb anything. It stays safe, dry, and perfectly solid. No high pressure outside = no melting! 🛡️✨
- **Why table salt gets wet in rain:** Table salt (sodium chloride) turns moist and ultimately forms a solution on exposure to air (especially during the rainy season). Though pure sodium chloride is not deliquescent, the commercial version of the salt contains impurities, like magnesium chloride and calcium chloride, which are deliquescent substances.
- **How to purify table salt:** This kind of impurity can be removed by passing a current of dry hydrogen chloride gas through a saturated solution of the affected salt. Pure sodium chloride is produced as a precipitate.
- **Why quicklime dries ammonia:** Quicklime being basic in nature is suitable for drying NH_3, a basic gas. Basic gas will not react with a base.
- **Proof of no water of crystallisation:** Both the substances (potassium chloride and sodium nitrate) do not contain water of crystallization as on heating them water vapours are not produced.

📊 3. Data & Tables Section

Table 3.5: Differences between drying agent and dehydrating agent ⚖️

Drying or desiccating agent	Dehydrating agent
(i) They remove moisture from other substances.	(i) They remove chemically combined elements of water in the ratio of 2 : 1 (hydrogen : oxygen) from a compound.
(ii) They are used to dry gases like chlorine, sulphur dioxide, hydrogen chloride, etc. They are also used in desiccators to keep substances dry.	(ii) They are used to prepare substances like carbon monoxide, sugar charcoal, etc.
(iii) They represent physical change as compositions do not change during drying of substances.	(iii) They represent chemical change as new substances are formed.
Examples: Phosphorus pentoxide P_2O_5, fused calcium chloride $CaCl_2$, calcium oxide CaO, conc. sulphuric acid H_2SO_4.	**Examples:** Conc. Sulphuric acid (H_2SO_4), alumina (Al_2O_3).

📌 **4. Characteristics & Exhaustive Lists Section**

- **Examples of crystalline substances which do not contain water of crystallisation:** 🚫💧
    - Sodium nitrate ($NaNO_3$)
    - Potassium chloride (KCl)
    - Lead chloride ($PbCl_2$)
    - Potassium sulphate (K_2SO_4)
    - Common salt ($NaCl$)
    - Nitre (KNO_3)
    - Sugar ($C_{12}H_{22}O_{11}$)
    - Potassium permanganate ($KMnO_4$)

- Ammonium chloride (NH_4Cl)

- **Examples of Efflorescent Substances:** 🌬️
  - Washing soda [hydrated sodium carbonate]
  - Glauber's salt [sodium sulphate decahydrate]
  - Epsom salt [magnesium sulphate heptahydrate]
- **Examples of Deliquescent Substances:** 🌧️
  - Caustic soda $NaOH$
  - Caustic potash KOH
  - Magnesium chloride $MgCl_2$
  - Zinc chloride $ZnCl_2$
  - Calcium chloride $CaCl_2$
  - Ferric chloride $FeCl_3$
- **Examples of Hygroscopic Substances:** 🧽
  - Conc. sulphuric acid (H_2SO_4) *(Note: Non-volatile)*
  - Phosphorus pentoxide (P_2O_5) *(Note: Non-volatile)*
  - Quicklime (CaO) *(Note: Non-volatile)*
  - Silica gel
  - *(Note: Conc. HCl and HNO_3 are volatile).*
- **How substances are dried:** 🏜️
  - **Gases** are dried by passing them through conc. sulphuric acid, a drying tower (quicklime), or a U-tube/drying bulb containing anhydrous sodium sulphate or anhydrous calcium chloride.
  - **Liquids** are dried by keeping them overnight over anhydrous Na_2SO_4 or $MgSO_4$ or $CaCl_2$ at room temperature. The solid is then removed by filtration.
  - **Solids** are dried by spreading them on a watch glass or a dish and keeping it in a desiccator for some days.

🧪 5. Reactions Master List Section

- **Heating Sodium Nitrate:**

$$2NaNO_3 \rightarrow 2NaNO_2 + O_2 \uparrow$$

- **Efflorescence of Washing Soda:**

$$Na_2CO_3 \cdot 10H_2O \xrightarrow{\text{dry air}} Na_2CO_3 \cdot H_2O + 9H_2O$$

- **Efflorescence of Glauber's Salt:**

$$Na_2SO_4 \cdot 10H_2O \xrightarrow{\text{dry air}} Na_2SO_4 + 10H_2O$$

- **Efflorescence of Epsom Salt:**

$$MgSO_4 \cdot 7H_2O \xrightarrow{\text{dry air}} MgSO_4 \cdot H_2O + 6H_2O$$

- **Dehydrating Blue Vitriol:**

$$CuSO_4 \cdot 5H_2O \xrightarrow{\text{Conc. } H_2SO_4} CuSO_4 + 5H_2O$$

- **Dehydrating Cane Sugar:**

$$C_{12}H_{22}O_{11} \xrightarrow{\text{Conc. } H_2SO_4} 12C + 11H_2O$$

- **Dehydrating Formic Acid:**

$$HCOOH \xrightarrow{\text{Conc. } H_2SO_4} CO + H_2O$$

Keep up the great work! ✨ Do you want to review anything else before your study time is up? ⏰

You are absolutely crushing it! 🔥 Let's keep this momentum going! Here are the exhaustive, 100% complete notes from the new pages you provided. I've extracted every single detail, reaction, and concept directly into our battle-tested format.

Let's dive right in! 🚀 📘

📖 1. Definitions & Core Concepts Section

- **Soft Water:** Water is said to be soft if it readily forms lather with soap. Pure water, distilled water, rain water, or water containing sodium salts are examples of soft water.

- **Hard Water:** Water is said to be hard when it does not readily form lather with soap. This is due to the presence of dissolved mineral matter, specifically hydrogen carbonates, sulphates, or chlorides of calcium and magnesium.
- **Temporary Hard Water:** Water that contains only hydrogen carbonates of calcium and magnesium. This hardness can be removed just by boiling.
- **Permanent Hard Water:** Water containing sulphates and chlorides of magnesium and calcium. This hardness cannot be removed by boiling.
- **Stalactite:** A conical pillar that grows downwards from the roof of a cave, formed by dripping water containing dissolved calcium hydrogen carbonate.
- **Stalagmite:** A pillar-like structure that grows upward from the floor of the cave. They can meet stalactites to form pillars.
- **Soap:** Chemically, soap is a sodium salt of stearic acid (an organic acid) with the formula $C_{17}H_{35}COONa$, represented as $NaSt$.
- **Soap-Curd or Scum:** A slimy, insoluble precipitate formed when calcium and magnesium ions in hard water combine with the negative ions of soap.
- **Synthetic Detergents:** "Soapless soaps" made by sulphonating higher alkenes (organic compounds) with sulphuric acid and converting them into sodium salts by caustic soda (e.g., SURF, DET).
- **Micelles:** Clusters of soap or detergent molecules in water, where their hydrocarbon tails stick inwards (attaching to oil/dirt) and negatively charged heads face outwards.

💡 2. The "Why" & Extra Key Facts Section

- **How water becomes hard:** When water flows over beds of limestone and dolomite ($CaCO_3, MgCO_3$), carbon dioxide slowly converts some of these into soluble calcium and magnesium bicarbonates, making it hard. Water flowing over gypsum ($CaSO_4 \cdot 2H_2O$) also gets hard by dissolving it.
- **Why stalactites form:** When water containing dissolved calcium hydrogen carbonate drips in caves, a release of pressure and warmth converts it back to solid calcium carbonate ($CaCO_3$).

- **Why hard water prevents lead poisoning:** Calcium sulphate in hard water forms an insoluble layer inside lead pipes, which checks lead poisoning. Soft water dissolves lead salts, making it poisonous.
- **Why hard water is bad for boilers:** Cold hard water enters hot boiler tubes, immediately turns to steam, and leaves behind dissolved solids as a deposit (boiler scale). Boiler scale is a bad conductor of heat, leading to fuel wastage, narrowed tubes, and dangerous pressure buildup that can cause the boiler to burst.
- **Why detergents are better than soap for hard water:** Detergents are unaffected by hardness because their calcium and magnesium salts are completely soluble in water, meaning they do not form scum.
- **How cleansing action works:** When water is stirred, oil and dirt lift off, breaking into fragments. Detergent tails stick to the dirt, while the negatively charged heads prevent the dirt globules from clumping back together, allowing them to be rinsed away.

📌 3. Characteristics & Exhaustive Lists Section

- **Advantages of Hard Water:**
    - **Tasty:** The presence of dissolved salts gives it a taste, making it ideal for the preparation of beverages and wines.
    - **Health:** Calcium and magnesium salts present are essential for the growth of bones and teeth.
    - **Safety:** Checks the poisoning of water by lead pipes.
- **Disadvantages of Hard Water:**
    - It causes corrosion in boilers and forms boiler scales (furring).
    - It causes massive wastage of soap because no lather forms until all Ca/Mg ions are precipitated as scum.
    - It gives a dull appearance to clothes and iron salts can form rusty spots.
    - It causes limescale buildup in pipes, water heaters, dishwashers, and washing machines, decreasing their efficiency.

- Mineral buildup corrodes pipe and plumbing systems, leading to leaks.
- Leaves stubborn mineral deposits on faucets, sinks, bathtubs, and toilets.
- Leaves a residue leading to dry and irritated skin, and dull and brittle hair.

- **Methods to Remove Temporary Hardness:**
  - **(1) By boiling:** Carbon dioxide is driven off, and soluble hydrogen carbonates convert into insoluble carbonates which are removed by filtration or decantation. *(Note: Not a practical method for large scale as it is costly and slow).*

🧪 4. Reactions Master List Section

- **Formation of Hard Water (Natural Process):**

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2 \text{ (soluble calcium bicarbonate)}$$

$$MgCO_3 + CO_2 + H_2O \rightarrow Mg(HCO_3)_2 \text{ (soluble magnesium bicarbonate)}$$

- **Formation of Stalactites / Stalagmites (And Removal of Temp Hardness by Boiling):**

$$Ca(HCO_3)_2 \xrightarrow{\text{Warmth/Pressure release/Boil}} CaCO_3 \downarrow + H_2O + CO_2 \uparrow$$

$$Mg(HCO_3)_2 \xrightarrow{\text{Boil}} MgCO_3 \downarrow + H_2O + CO_2 \uparrow$$

- **Soap Reacting with Temporary Hard Water (Forming Scum):**

$$2NaSt \text{ (soap)} + Ca(HCO_3)_2 \rightarrow CaSt_2 \downarrow \text{ (soap curd)} + 2NaHCO_3$$

$$2NaSt + Mg(HCO_3)_2 \rightarrow MgSt_2 \downarrow + 2NaHCO_3$$

- **Soap Reacting with Permanent Hard Water (Forming Scum):**

$$2NaSt + CaCl_2 \rightarrow CaSt_2 \downarrow + 2NaCl$$

$$2NaSt + MgCl_2 \rightarrow MgSt_2 \downarrow + 2NaCl$$

$$2NaSt + CaSO_4 \rightarrow CaSt_2 \downarrow + 2Na_2SO_4$$

$$2NaSt + MgSO_4 \rightarrow MgSt_2 \downarrow + 2Na_2SO_4$$

📖 1. Definitions & Core Concepts Section

- **Potable Water:** Water that is safe enough for human consumption without risk of immediate or long-term illness.
- **Permutit:** Chemically, it is an artificial zeolite, specifically a hydrated sodium aluminium orthosilicate with the formula $Na_2Al_2Si_2O_8 \cdot XH_2O$ (simplified as Na_2P).
- **Zeolite:** A naturally formed mineral found in abundance in large deposits of volcano tuff that can easily be dried and used for ion exchange.
- **Coagulants (e.g., Alum):** Chemicals mixed with raw untreated water to help purify it. Alum is a white crystalline hydrated double sulphate salt used for centuries.
- **Flocs:** Tiny sticky globs formed when alum is added to water. Bacteria, mud, and other impurities stick to these flocs and settle to the bottom.
- **Chlorination:** A process in which chlorine is added to water (acting as a disinfectant) to kill harmful microorganisms and destroy disease-causing bacteria.
- **Defluoridation:** The process of removing excess fluoride from drinking water to achieve safe fluoride levels and prevent health problems.

💡 2. The "Why" & Extra Key Facts Section

- **Why Clark's Process (lime) is used:** It is a practical method to remove temporary hardness by adding slaked lime, which causes calcium and magnesium carbonates to precipitate so they can be filtered out.
- **Why Permutit needs regeneration:** After the sodium ions in the permutit are completely exchanged for calcium and magnesium ions from the hard water, it becomes inactive. It is regenerated by running a concentrated solution of brine (NaCl) over it and washing away the formed calcium chloride.
- **Why scientific work requires deionised water:** For scientific purposes, even the presence of positive sodium ions and negative bicarbonate and sulphate ions is not desirable. Synthetic ion exchange resins replace

sodium ions with H^+ ions and negative ions with OH^- ions to completely deionise the water.

- **Why defluoridation is necessary:** High levels of fluoride in water can lead to severe health problems like dental and skeletal fluorosis (tooth and bone disorders).
- **Why water is aerated:** Aeration improves the overall taste and odour of drinking water.
- **Why chemical contamination is so dangerous:** Long-term consumption of water chemically contaminated with industrial wastes, pesticides, or heavy metals can increase the risk of cancer and cause severe organ damage, particularly to the liver, kidneys, and other vital organs.

📊 3. Data & Tables Section

Characteristics of Potable Water 💧

Property	Standard Requirement
Appearance	Clear and transparent
pH	Between 6.5 and 8.5
TDS (Total Dissolved Solids)	Should not exceed about 2000 mg/L
Total Hardness	Generally within 300-600 mg/L
Chlorides	Within 250-1000 mg/L
Fluoride	Safe range 1.0-1.5 mg/L
Iron	Within 0.3-1.0 mg/L

📌 4. Characteristics & Exhaustive Lists Section

- **Steps for Making Water Potable (in order):**

1. Addition of Coagulants.
2. Coagulation and Settling (Aeration and sedimentation).
3. Filtration (using layers of sand, gravel, and coal).
4. Disinfection / Chlorination.
5. Final treatment (e.g., pH Correction, UV light) before distribution.

- **Causes of Drinking Water Contamination:**
  - Sewage improperly disposed of, leaking into soil and underground supplies.
  - Open defecation or urination near lakes and rivers.
  - Animal wastes draining from dairies and poultry farms.
  - Industrial wastes containing toxic chemicals.
  - Pesticides and chemical fertilisers washing from agricultural fields.
- **Common Waterborne Diseases:**
  - **Bacterial:** Diarrhoea, typhoid, cholera, and dysentery (these affect the digestive system causing stomach pain, fever, vomiting, dehydration).
  - **Viral:** Hepatitis A (affects the liver causing fever/weakness) and Polio (affects the nervous system and may lead to paralysis).
- **Common Defluoridation Techniques:**
  - Adsorption (using activated alumina or charcoal).
  - Chemical precipitation (using alum).
  - Membrane processes (like reverse osmosis (RO) and electrodialysis).

🧪 5. Reactions Master List Section

- **Removal of Temporary Hardness via Clark's Process (Addition of Slaked Lime):**

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + 2H_2O$$

$$Mg(HCO_3)_2 + Ca(OH)_2 \rightarrow MgCO_3 \downarrow + CaCO_3 \downarrow + 2H_2O$$

- **Removal of Temporary Hardness via Washing Soda:**

$$Ca(HCO_3)_2 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + 2NaHCO_3$$

$$Mg(HCO_3)_2 + Na_2CO_3 \rightarrow MgCO_3 \downarrow + 2NaHCO_3$$

- **Removal of Permanent Hardness via Washing Soda (Soda Ash):**

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + Na_2SO_4$$

$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + 2NaCl$$

- **Regeneration of Permutit (after it becomes inactive):**

$$CaP + 2NaCl \rightarrow Na_2P + CaCl_2$$

EXERCISE 3A

1. **Water exists in all the three states. Discuss.**

Water naturally exists as a solid (ice) below 0°C, as a liquid (water) between 0°C and 100°C, and as a gas (steam or water vapor) above 100°C. 🧊 💧 ☁️

2. **Why is water considered a compound?**

Water is made of two different elements, Hydrogen and Oxygen, chemically combined in a fixed ratio. You cannot separate them using simple physical methods, which makes it a chemical compound. 🧪 💧

3. **(a) Why does temperature in Mumbai and Chennai not fall as low as it does in Delhi?**

Mumbai and Chennai are near the sea. Water has a very high specific heat capacity, meaning it absorbs and releases heat very slowly, keeping the coastal climate moderate. Delhi is far from the sea, so it experiences extreme temperatures. 🌊 ☀️

(b) Give the properties of water responsible for controlling the temperature of our body.

Water's high specific heat capacity and high latent heat of vaporization help our body absorb heat without our internal temperature spiking, and sweating cools us down fast. 🏃 💦

4. **'Water is a universal solvent'. Comment.**

Water can dissolve a massive variety of substances, both solid salts and gases, better than almost any other liquid on Earth. That is exactly why it is called a universal solvent. 🌍💧

5. **What causes the violence associated with torrential rain?**

When water vapor condenses into rain clouds, it releases a massive amount of hidden heat (latent heat of condensation). This sudden heat warms the surrounding air, making it expand and rush upward rapidly, which creates violent winds and storms. 🌧️🌪️

6. **(a) Which property of water enables it to modify the climate?**

Its remarkably high specific heat capacity allows it to absorb and release large amounts of heat slowly. 🌡️🌊

(b) Density of water varies with temperature. What are its consequences?

Water is heaviest (most dense) at exactly 4°C. Because of this, ice floats on top of liquid water, and lakes freeze from the top down instead of bottom up. 🧊⬆️

(c) What is the effect of impurities present in water on melting point and boiling point of water?

Adding impurities raises the boiling point (water boils hotter than 100°C) and lowers the freezing point (water needs to be colder than 0°C to freeze). 🧂🔥

7. **How do fishes and aquatic animals survive in winters when the pond gets covered with thick ice?**

Because water is heaviest at 4°C, this slightly warm water sinks to the very bottom of the pond. The colder 0°C water stays at the top and freezes into ice. The ice acts like an insulating blanket, keeping the liquid water at the bottom safe for fish to live in. 🐠❄️

8. **The properties of water are different from the properties of the elements of which it is formed. Discuss.**

Hydrogen is a gas that catches fire, and Oxygen is a gas that helps things burn. But when they combine into water, they form a liquid that puts out fires. This shows the compound has totally different properties than its ingredients. 🔥🧯

9. **How is aquatic life benefitted by the fact that water has maximum density at 4°C?**

The densest water sinks to the bottom, meaning the bottom of a lake won't freeze even in winter. This gives aquatic life a liquid, safe zone to survive. 🦞🥶

10. **What are your observations and conclusion when tap water is boiled and evaporated in watch glass?**

Observation: You will see faint, white circular rings left behind on the glass. 👀

Conclusion: Tap water contains dissolved solid impurities and salts. 🚰🧂

11. **What is the importance of dissolved salts in water?**

Dissolved salts give drinking water its refreshing taste, and they provide essential minerals like calcium that our bodies need for strong bones and teeth. 😋🦴

12. **State the importance of the solubility of CO2 and O2 in water.**

Dissolved Oxygen (O2) is essential for fish and marine animals to breathe. Dissolved Carbon Dioxide (CO2) is needed by underwater plants to make their food via photosynthesis. 🌿🐟

13. **How is air dissolved in water different from ordinary air?**

Air that is dissolved in water actually has a much higher percentage of oxygen (about 33%) than the normal air we breathe (21%). This is because oxygen dissolves into water much more easily than nitrogen does. 🫧📈

14. **Identify A, B, C and D; first one is done for you.**

(A) Latent heat of fusion (Solid melting to Liquid) 🧊➡️💧

(B) Latent heat of vaporization (Liquid boiling to Gas) 💧➡️☁️

(C) Condensation (Gas cooling to Liquid) ☁️➡️💧

(D) Freezing / Solidification (Liquid freezing to Solid) 💧➡️🧊

15. **Explain why:**

(a) Boiled or distilled water tastes flat.

Boiling forces all the dissolved gases out of the water, and those gases are what give water its good taste. ♨️👅

(b) Ice at zero degree centigrade has greater cooling effect than water at 0°C.

Ice absorbs extra hidden heat (latent heat of fusion) from its surroundings just to melt into water, so it pulls much more heat away than cold water does. 🧊🥶

(c) Burns caused by steam are more severe than burns caused by boiling water.

Steam holds a massive amount of extra heat energy (latent heat of vaporization) compared to boiling water, so it transfers way more damaging heat to your skin. 🔥🤯

(d) Rivers and lakes do not freeze easily.

Water takes a very long time to lose its heat because of its high specific heat capacity, so it needs prolonged, extreme cold to freeze over. 🏞️⏳

(e) Air dissolved in water contains a higher proportion of oxygen.

Oxygen is simply much more soluble in water than nitrogen is. 🍽️🌊

(f) If distilled water is kept in a sealed bottle for a long time, it leaves etchings on the surface of the glass.

Distilled water is such a powerful solvent that over a long time, it starts to slowly dissolve the glass itself, making it cloudy or etched. 🫙🔍

(g) Rain water does not leave behind concentric rings when boiled.

Rain water is a naturally pure form of water, meaning it doesn't have any dissolved solid salts to leave behind as a residue. 🌧️✨

EXERCISE 3B

1. How will you justify that air is a mixture and not an element or compound?

Air is considered a mixture because its components (oxygen, nitrogen, etc.) retain their individual properties and can be separated by physical methods like fractional distillation. Also, its composition varies slightly depending on where you are! 💨🌍

2. Hydrogen chloride gas is a _____ (pure/impure) substance, but when it is dissolved in water then it is a _____ (homogeneous/heterogeneous) mixture.

pure, homogeneous. 💉💧

3. Rani got a cut on her finger while working in the kitchen. She applied an antiseptic liquid to stop bleeding but it was not effective. Her husband asked her to rub alum on the cut. To her surprise, the bleeding stopped.

(a) Why was the antiseptic liquid not effective?

Antiseptics are meant to kill germs and prevent infection, not to clot blood. 🦠🩸

(b) What was the use of alum here?

Alum acts as a coagulating agent. Blood is a colloidal solution, and alum causes the blood particles to clump together and form a clot, stopping the bleeding. 🛑🩹

(c) Who according to you has a proper knowledge of colloidal solutions — Rani or her husband?

Her husband! 👨‍🔬🧠

4. When a beam of light was passed through the solution of a substance 'X' dissolved in water, the path of light could be seen. What is the nature of this solution?

It is a colloidal solution because it exhibits the Tyndall effect (scattering of light). 🔦🌫️

5. Give an example each for a mixture having the following characteristics:

(a) Two or more coloured constituents soluble in a solvent.

Black ink (contains different coloured dyes dissolved in water). 🖋️🌈

(b) Two immiscible liquids.

Oil and water. 🛢️💧

6. Identify colloids from the following: Copper sulphate solution, milk, smoke, muddy water, butter, sugar solution, face cream, lemonade, jelly, soda water.

Milk, smoke, butter, face cream, and jelly. 🥛🧈

7. Identify the dispersed phase and dispersion medium in the following examples of colloids:

(a) Fog

Dispersed phase: Liquid, Dispersion medium: Gas. 🌫️

(b) Cheese

Dispersed phase: Liquid, Dispersion medium: Solid. 🧀

(c) Coloured gem stones

Dispersed phase: Solid, Dispersion medium: Solid. 💎

8. Which of the following solutions exhibit Tyndall effects, starch solution, sodium chloride solution, tincture of iodine, air.

Starch solution (because it is a colloid). ✨🥚

9. List three differences between true solutions and suspensions.

10. True solutions are perfectly homogeneous, while suspensions are heterogeneous. 🥣

11. Particles in a true solution are incredibly tiny (less than 1 nm), while suspension particles are large and visible (greater than 1000 nm). 📏

12. True solutions do not scatter light, but suspensions do scatter light (until the particles settle). ✨

13. Name the type of colloid in each of the following, giving an example of each. (Dispersed phase / Dispersion medium: X: Liquid/Gas, Y: Liquid/Liquid, Z: Liquid/Solid)

X (Liquid in Gas): Aerosol (Example: Fog or mist). 🌫️

Y (Liquid in Liquid): Emulsion (Example: Milk or face cream). 🥛

Z (Liquid in Solid): Gel (Example: Jelly or butter). 🍮

14. How are colloids different from true solutions?

Colloids are heterogeneous mixtures (though they look homogeneous) with medium-sized particles that scatter light (Tyndall effect). True solutions are strictly homogeneous mixtures with tiny particles that dissolve completely and do not scatter light. 🔦🔬

15. Compare suspensions and colloidal solutions on the basis of:

(a) type of mixture

Suspension is visibly heterogeneous; Colloid is heterogeneous but looks homogeneous. 👀

(b) particle size

Suspension has large particles (> 1000 nm); Colloid has medium particles (1 nm to 1000 nm). 📏

(c) scattering of light

Suspension scatters light but only until the heavy particles settle; Colloid continuously scatters light perfectly (Tyndall effect). 🔦

(d) stability

Suspension is very unstable (particles settle down); Colloid is quite stable (particles do not settle). ⚖️

16. What do you understand by 'Tyndall effect'? Which of the following will show it? (a) Salt solution (b) milk (c) copper sulphate solution (d) starch solution.

The Tyndall effect is the scattering of a beam of light by colloidal particles, which makes the path of the light visible. Both (b) milk and (d) starch solution will show it! 🔦🥛

17. Explain the terms:

(a) solution

A perfectly uniform, homogeneous mixture of two or more substances. 🧃

(b) solute

The substance that gets dissolved (like the salt in salt water). 🧂

(c) solvent

The liquid that does the dissolving (like the water in salt water). 💧

18. Explain why a hot saturated solution of potassium nitrate forms crystals as it cools.

As the temperature drops, the solubility of potassium nitrate decreases. The extra dissolved solid can no longer stay mixed in the cooler liquid, so it separates out as beautiful solid crystals. ❄️💎

19. Give three factors which affect the solubility of a solid solute in a solvent.

20. Temperature of the solvent. 🌡️

21. The nature of the solute and the solvent. 🧪

22. The surface area (size) of the solute particles. 🧊

23. (a) If you are given some copper sulphate crystals, how would you proceed to prepare its saturated solution at room temperature?

Keep adding the crystals to a fixed amount of water at room temperature and stir continuously. Stop when no more crystals can dissolve and they start settling at the bottom. 🥄💧

(b) How can you show that your solution is really saturated?

Drop one more tiny crystal into the beaker. If it doesn't dissolve at all and just sinks to the bottom, the solution is definitely saturated! 🛑💎

24. What is the effect of temperature on solubility of KNO_3 and $CaSO_4$ in water?

For KNO_3, solubility increases rapidly as the temperature goes up. For $CaSO_4$, its solubility actually decreases slightly as the temperature goes up. 📈📉

25. Solubility of NaCl at 40°C is 36.5 g. What is meant by this statement?

It means that exactly 36.5 grams of NaCl (salt) can dissolve in 100 grams of water at 40°C to create a perfectly saturated solution. ⚖️🧂

26. Which test will you carry out to find out if a given solution is saturated or unsaturated or supersaturated?

Add a small pinch of the solute to the solution. If it dissolves, the solution is unsaturated. If it just sinks to the bottom undissolved, it is saturated. If adding it causes rapid, sudden crystallization of extra solid, it was supersaturated! 🧪🔍

27. What happens when a saturated solution of nitre (KNO_3) prepared at 60°C is allowed to cool at room temperature?

As it cools down to room temperature, the solubility of nitre drops sharply. The extra dissolved nitre will fall out of the solution and form visible solid crystals at the bottom of the container. 🧊📉

EXERCISE 3C

1. (a) Define (i) Henry's law (ii) Crystallization (iii) Seeding.

(i) **Henry's Law:** At a constant temperature, the mass of a gas dissolved in a given volume of a liquid is directly proportional to the pressure of the gas on the surface of the liquid.

(ii) **Crystallization:** The process by which solid crystals of a substance separate out from a hot saturated solution as it cools.

(iii) **Seeding:** The process of inducing crystallization in a supersaturated solution by introducing a small, pure crystal (the "seed") of the same substance.

(b) State any three methods of crystallization.

1. Cooling a hot saturated solution.
2. Slow evaporation of a saturated solution at room temperature.
3. Cooling a fused (melted) mass of the substance.

2. What would you observe when crystals of copper (II) sulphate are heated in a test tube strongly?

The blue crystals of copper (II) sulphate pentahydrate ($CuSO_4 \cdot 5H_2O$) lose their water of crystallization and crumble into a powdery, white mass of anhydrous copper (II) sulphate ($CuSO_4$). Additionally, water droplets condense on the cooler, upper inner walls of the test tube.

3. Give the names and formulae of two substances in each case

(a) hydrated substance:

1. Copper (II) sulphate pentahydrate / Blue vitriol ($CuSO_4 \cdot 5H_2O$)
2. Sodium carbonate decahydrate / Washing soda ($Na_2CO_3 \cdot 10H_2O$)

(b) anhydrous substance:

3. Anhydrous calcium chloride ($CaCl_2$)
4. Sodium chloride ($NaCl$)

(c) liquid drying agent:

5. Concentrated sulphuric acid (H_2SO_4)

(d) a basic drying agent:

6. Calcium oxide / Quicklime (CaO)
7. Soda lime (mixture of $NaOH$ and CaO)

4. What is the effect of pressure on solubility of gases. Explain with an example.

The solubility of a gas in a liquid increases with an increase in pressure (Henry's Law).

Example: Soda water contains carbon dioxide gas dissolved in water under high pressure. When the bottle is opened, the pressure drops abruptly, causing the solubility of CO_2 to decrease, and the excess gas rapidly escapes as bubbles (effervescence).

5. State the term : (Do not give examples)

(a) A solution where solvent is a liquid other than water.

Non-aqueous solution.

(b) When a substance absorbs moisture on exposure to moist air and dissolves in the absorbed water and turned to solution.

Deliquescence.

(c) A substance which contains water of crystallisation.

Hydrated salt / Hydrate.

(d) When a substance absorbs moisture from the atmosphere, but does not form a solution.

Hygroscopy.

(e) When a compound loses its water of crystallisation on exposure to dry air.

Efflorescence.

(f) The substance that can remove hydrogen and oxygen atoms in the ratio of 2 : 1 (in the form of water) from the compounds.

Dehydrating agent.

6. Complete the following table :

Common Name	**Chemical Name**	**Formula**	**Acid, base or salt**	**Efflorescent, hygroscopic or deliquescent substance**
Solid caustic potash	Potassium hydroxide	KOH	Base	Deliquescent
Quick lime	Calcium oxide	CaO	Base	Hygroscopic
Oil of vitriol	Concentrated sulphuric acid	H_2SO_4	Acid	Hygroscopic
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3 \cdot 10H_2O$	Salt	Efflorescent

Common Name	Chemical Name	Formula	Acid, base or salt	Efflorescent, hygroscopic or deliquescent substance
Solid caustic soda	Sodium hydroxide	$NaOH$	Base	Deliquescent
Blue vitriol	Copper(II) sulphate pentahydrate	$CuSO_4 \cdot 5H_2O$	Salt	Efflorescent (mildly, in highly dry air)

7. Explain why :

(a) water is an excellent liquid to use in cooling systems.

Water has a remarkably high specific heat capacity, meaning it can absorb large amounts of heat energy with only a slight increase in its own temperature.

(b) a solution is always clear and transparent.

In a true solution, the solute particles are broken down to the molecular or ionic level (less than 1 nm in size). They are uniformly distributed and are too small to scatter light, making the mixture clear.

(c) lakes and rivers do not suddenly freeze in the winters.

Water has a high specific heat capacity, so it takes a very long time to lose enough heat to freeze. Additionally, its anomalous expansion means water is densest at $4°C$; this dense water sinks, leaving the top layer to freeze first, which acts as an insulator for the liquid water below.

(d) the solute cannot be separated from a solution by filtration.

The solute particles in a solution are exceptionally small and pass directly through the microscopic pores of filter paper along with the solvent.

(e) fused $CaCl_2$ or conc. H_2SO_4 is used in a desiccator.

Both are powerful hygroscopic substances (drying agents). They rapidly absorb moisture from the enclosed air, keeping the environment inside the desiccator completely dry.

(f) effervescence is seen on opening a bottle of soda water.

Soda water is bottled under high pressure to force CO_2 to dissolve. Opening the bottle reduces the pressure to atmospheric levels. According to Henry's Law, the gas's solubility decreases, forcing the excess CO_2 to bubble out rapidly.

(g) Table salt becomes sticky on exposure to humid air during the rainy season.

Commercial table salt contains tiny amounts of impurities like magnesium chloride ($MgCl_2$) and calcium chloride ($CaCl_2$). These impurities are highly deliquescent and absorb moisture from the humid air, making the salt sticky.

8. Normally, solubility of a crystalline solid increases with temperature. Does it increase uniformly in all cases ? Name a substance whose solubility :

No, solubility does not increase uniformly in all cases; it depends on the specific substance.

(a) increases rapidly with temperature.

Potassium nitrate (KNO_3).

(b) increases gradually with temperature.

Potassium chloride (KCl).

(c) increases slightly with temperature.

Sodium chloride ($NaCl$).

(d) initially increases then decreases with rise in temperature.

Sodium sulphate decahydrate / Glauber's salt ($Na_2SO_4 \cdot 10H_2O$).

9. What are drying or desiccating agents. Give examples.

Drying agents are substances that can absorb moisture from other substances or from the surrounding air without reacting chemically with them. They are hygroscopic in nature.

Examples: Anhydrous calcium chloride ($CaCl_2$), Quicklime (CaO), Concentrated sulphuric acid (H_2SO_4).

10. In which of the following substances will there be : (a) increase in mass (b) decrease in mass (c) no change in mass when they are exposed to air ?

1. **Sodium chloride (pure):** (c) no change in mass.
2. **Iron:** (a) increase in mass (due to absorption of oxygen and moisture to form rust).
3. **Conc. sulphuric acid:** (a) increase in mass (it is hygroscopic and absorbs moisture from the air).
4. **Table salt (commercial):** (a) increase in mass (due to deliquescent impurities like $MgCl_2$ absorbing water).
5. **Sodium carbonate crystals:** (b) decrease in mass (it is efflorescent and loses its water of crystallization to the air).

11. State the methods by which hydrated salts can be made anhydrous ?

1. **Direct Heating:** Strongly heating the hydrated salt until all water of crystallization is driven off as steam.
2. **Desiccation:** Placing the hydrated salt inside a desiccator containing a strong drying agent (like concentrated H_2SO_4 or anhydrous $CaCl_2$) for an extended period, which draws the moisture out of the salt.

EXERCISE 3D

(A) MULTIPLE CHOICE TYPE

(i) Sodium sulphate is soluble in:

(b) Water 💧

(ii) Water acts as a universal solvent because:

(b) It is polar and has a high dielectric constant. ⚡

(iii) Permanent hardness of water is removed by:

(d) Adding sodium carbonate. 🧪

(iv) Solid solutions are called:

(c) Alloys 🛡️

(v) With a rise in temperature, the solubility of sodium chloride will:

(d) Increase slightly 📈

(vi) The solubility of which of the following substances decreases with a rise in temperature:

(c) Calcium hydroxide 📉

(vii) The crystalline substance that does not contain water of crystallisation is:

(c) Potassium permanganate 💜

(viii) Table salt becomes sticky in the presence of moisture because:

(c) It contains impurities which are deliquescent 🧂

(ix) Which of the following substances can act as a dehydrating as well as a drying agent?

(d) Conc. sulphuric acid 🧪

(x) Which of the following is a deliquescent salt?

(b) $FeCl_3$ 💦

(xi) If a salt on heating does not give water vapour, then that salt is:

(d) Anhydrous 🏜️

(xii) The salt which is the cause of hardness in water is:

(b) Magnesium bicarbonate 🪨

(xiii) Temporary hardness of water can be removed by:

(b) Boiling ♨️

(xiv) The salt which does not contain any water of crystallisation is: P - Blue vitriol, Q - Gypsum, R - Baking soda

(c) Only R 🧁

(xv) Sodium chloride (common salt) besides being used in kitchen can also be used as the raw material for making: P - Slaked lime, Q - Washing soda, R - Baking soda

(d) Both Q and R 🫧

(xvi) The figure shown below demonstrates the solubility curve of a substance. From the statements given below, choose which is/are correct: P - A is hydrated sodium chloride, B is anhydrous sodium chloride. Q - A is Glauber's salt, B is sodium sulphate. R - A is Gypsum, B is calcium sulphate.

(b) Only Q 📊

(B) VERY SHORT ANSWER TYPE

1. Name:

(a) The solute and solvent in sugar solution.

Solute: Sugar. Solvent: Water. 🍖 💧

(b) The characteristic property which makes water the universal solvent.

Its polar nature and high dielectric constant. ⚡

(c) A substance whose solubility shows an anomalous behaviour.

Glauber's salt (Sodium sulphate decahydrate). 📉 📈

(d) A substance whose solubility rapidly increases with the temperature.

Potassium nitrate (KNO_3). 🚀

2. What is the composition of water? In what volume does its elements combine?

Water is composed of Hydrogen and Oxygen. They combine in a volume ratio of 2:1. 🧪

3. Name the substances which give water:

(a) temporary hardness

Bicarbonates of calcium and magnesium. 🪨

(b) permanent hardness.

Chlorides and sulphates of calcium and magnesium. 🧱

4. Name the three methods by which hydrous substances can be made anhydrous.

1. Direct heating. 2. Desiccation (using strong drying agents like conc. H_2SO_4). 3. Exposing efflorescent salts to dry air. 🔥 🧂

(C) SHORT ANSWER TYPE

1. What is the use of solubility of oxygen and carbon dioxide in water?

Dissolved oxygen is essential for the survival and respiration of aquatic animals like fish. Dissolved carbon dioxide is used by aquatic plants for photosynthesis. 🐟 🌿

2. Hot saturated solution of sodium nitrate forms crystals, as it cools. Why?

Because the solubility of sodium nitrate decreases as the temperature falls. The cooler water cannot hold all the dissolved solute, forcing the excess to separate out as solid crystals. ❄️ 💎

3. What do you understand by

(a) Soft water

Water that easily produces a rich lather with soap. 🫧

(b) Hard water

Water that does not lather easily with soap, instead forming a sticky, white scum. 🧼 🚫

(c) Temporary hard water

Hardness that can be easily removed just by boiling the water. ♨️

(d) Permanent hard water.

Hardness that cannot be removed by simply boiling; it requires chemical treatment. 🧪

4. What are the causes for

(a) Temporary hardness

Presence of dissolved calcium and magnesium bicarbonates. 🪨

(b) Permanent hardness

Presence of dissolved calcium and magnesium chlorides and sulphates. 🧱

5. State two ways by which a saturated solution can be changed to unsaturated solution.

1. By increasing the temperature (heating it). 2. By adding more solvent (water) to it. 🔥 💧

6. What is a soap, what is it used for?

A soap is a sodium or potassium salt of higher fatty acids. It is used for washing and cleaning purposes. 🧼 👕

7. What is the advantage of a detergent over soap?

Detergents can easily form a good lather even in hard water, whereas soaps fail to lather and wastefully form scum. 🫧 💪

8. Explain why water that is chemically pure is not necessarily 'potable water'.

Chemically pure water (like distilled water) has no dissolved gases or minerals. Therefore, it tastes completely flat and lacks the essential minerals required for our body's health, making it unfit for everyday drinking (potable). 👅 🚫

9. Even if water looks clear and has no odour why it might still be unsafe to drink?

It might contain dissolved toxic chemicals, heavy metals, or microscopic disease-causing pathogens (bacteria/viruses) that are completely invisible to the naked eye. 🦠 ☠️

10. Why should we not allow our household drainage water to flow into nearby water bodies?

It causes severe water pollution, depletes dissolved oxygen (harming aquatic life), and spreads dangerous water-borne diseases. 🚫 🌊

11. Name three water-borne diseases that can be prevented by drinking potable water.

Cholera, Typhoid, and Dysentery. 🤢🛡️

12. Residents of a certain village rely on a local river for drinking water. What are the potential health risks that they might face?

They are at a high risk of contracting water-borne infectious diseases (like cholera or typhoid) from biological contamination, and poisoning from agricultural/industrial chemical runoffs. ⚠️🏥

(D) DESCRIPTIVE TYPE

1. What are hydrous substances? Explain with examples.

Hydrous substances are crystalline salts that contain a fixed, definite number of water molecules chemically attached to them (water of crystallization). Example: Blue vitriol ($CuSO_4 \cdot 5H_2O$) and Washing soda ($Na_2CO_3 \cdot 10H_2O$). 💎💧

2. What is the importance of dissolved impurities in water?

Dissolved impurities (like natural salts and minerals) give drinking water its pleasant taste and supply our bodies with essential minerals like calcium and magnesium for bone health. 😋🦴

3. What are the advantages of (i) soft water (ii) hard water?

(i) Soft water: Lathers easily, saving soap and making washing clothes much more efficient. 🫧👕

(ii) Hard water: Provides essential minerals for strong teeth and bones, and is highly preferred in the brewing industry for making beverages. 🍺🦴

4. What are stalagmites and stalactites? How are they formed?

They are cone-shaped mineral deposits found in limestone caves. They form when water containing dissolved calcium bicarbonate drips from the cave roof. As the water evaporates and loses carbon dioxide, it leaves behind solid calcium carbonate. Stalactites hang from the ceiling, and stalagmites grow from the floor! 🦇🌰

5. Give equations to show what happens when temporary hard water is (a) boiled, (b) treated with slaked lime.

(a) Boiled:

$Ca(HCO_3)_2 \xrightarrow{\text{heat}} CaCO_3 \downarrow + H_2O + CO_2 \uparrow$ 🔥

(b) Treated with slaked lime:

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + 2H_2O$ 🧪

6. State the disadvantages of using hard water.

It wastes a huge amount of soap, ruins fabrics by depositing sticky scum, makes water unfit for dyeing, and forms dangerous scales inside industrial boilers. 🚫🧼

7. Why does the hardness of water render it unfit for use in a (i) boiler (ii) for washing purposes?

(i) Boiler: It forms a hard, stony crust called "boiler scale" on the inside. This scale wastes fuel because it blocks heat, and can eventually cause the boiler to explode! 💥🔥

(ii) Washing: It reacts with soap to form insoluble sticky scum instead of lather, meaning clothes don't get clean and soap is wasted. 👕👎

8. Explain with equation, what is noticed when permanent hard water is treated with (a) slaked lime (b) washing soda

(Note: Slaked lime is generally for temporary hardness. For permanent hardness, washing soda is the standard chemical).

(b) Treated with washing soda: The soluble calcium/magnesium sulphates react with washing soda to form insoluble carbonates that precipitate out, making the water soft.

Equation: $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + Na_2SO_4$ 🧪

9. What is permutit method, how can it be used for softening hard water?

Permutit is an artificial zeolite (hydrated sodium aluminium silicate). When hard water is passed through a tall cylinder packed with permutit, it undergoes "ion exchange." The calcium and magnesium ions in the water swap places with the sodium ions in the permutit, completely softening the water. 🔄💧

10. Explain, with equations, why ordinary soap does not lather easily in hard water.

Soap (Sodium stearate) reacts chemically with the calcium or magnesium ions present in hard water to form an insoluble precipitate called scum (Calcium stearate), rather than lathering.

Equation: $2C_{17}H_{35}COONa + Ca^{2+} \rightarrow (C_{17}H_{35}COO)_2Ca \downarrow +2Na^+$ 🧼📉

11. Explain:

(a) The use of lead pipes for drinking water supply is being discontinued.

Water slightly dissolves lead (a process called plumbo-solvency), which accumulates in the human body and causes severe lead poisoning. ☠️🚰

(b) Chalk hills dissolve in rain water.

Rainwater mixes with atmospheric carbon dioxide to form weak carbonic acid. This acidic water reacts with the insoluble chalk (calcium carbonate) hills, converting them into soluble calcium bicarbonate, literally dissolving the rock! 🌧️🟫

(c) Hard water is unfit for boilers.

Because the dissolved salts precipitate out on heating to form a rock-hard layer of boiler scale, which wastes fuel and risks deadly explosions. 🏭💥

(d) Iron chloride forms a saturated solution when exposed to the atmosphere.

Iron (III) chloride is highly deliquescent. It absorbs so much water vapor from the air that it eventually dissolves in its own absorbed moisture to form a saturated solution. 💦🧪

(e) A bottle containing concentrated H_2SO_4 should be stoppered.

Concentrated sulphuric acid is intensely hygroscopic. If left open, it will rapidly absorb moisture from the air, diluting itself and increasing in volume until it dangerously overflows. 🛑☢️

(E) STRUCTURED/APPLICATION/SKILL TYPE

1. Some salts while crystallising out from their solutions, combine with a definite quantity of water. Some of these salts lose this water on exposure to dry air. Certain water soluble substances, when exposed to atmosphere

at ordinary temperatures, absorb moisture from atmospheric air forming a saturated solution.

(a) State the term used for a 'definite quantity of water combined with a salt.

Water of crystallization. 💎💧

(b) What type of salts lose their water molecules? Give two examples.

Efflorescent salts. Examples: Washing soda ($Na_2CO_3 \cdot 10H_2O$) and Glauber's salt ($Na_2SO_4 \cdot 10H_2O$). 🏜️✨

(c) Table salt forms a solution on exposure to air during rainy season. Explain.

Commercial table salt contains tiny amounts of impurities like magnesium chloride ($MgCl_2$) and calcium chloride ($CaCl_2$). These specific impurities are highly deliquescent and absorb the heavy atmospheric moisture during the rainy season, making the salt wet. 🌧️🧂

(d) How can pure sodium chloride be obtained from the above table salt solution in (c)?

By passing dry hydrogen chloride (HCl) gas through a saturated solution of the impure commercial salt. Due to the common ion effect, pure sodium chloride crystals will precipitate out, leaving the impurities dissolved. 🧪💎

2. The following figure shows the solubility curves of $NaCl$, KNO_3 and hydrated calcium sulphate.

(a) Identify and label the curves with the salt it represents.

The curve shooting rapidly upwards is KNO_3. The flat, almost straight horizontal line is $NaCl$. The curve sloping downwards is hydrated calcium sulphate. 📈📉

(b) State the factors on which the solubility depends.

Temperature, the nature of the solute, and the nature of the solvent. 🌡️🧪

(c) Solubility of which salt(s) shows:

(i) Endothermic process

KNO_3 and $NaCl$ (because their solubility increases as heat is added). 🔥

(ii) Exothermic process?

Hydrated calcium sulphate (because its solubility decreases as heat is added). ❄️

(F) ASSERTION - REASON TYPE QUESTIONS

(i) Assertion (A) : Water is a universal solvent. Reason (R) : Water dissolves all substances except noble metals and glass.

(c) A is true but R is false. (Water dissolves many things, but definitely not *all* other substances!) 🌍

(ii) Assertion (A) : A saturated solution becomes unsaturated on heating. Reason (R) : More amount of solute can dissolve in a solvent upon heating.

(a) Both A and R are true and R is the correct explanation of A. 🔥📈

(iii) Assertion (A) : A white powder forms on the surface of washing soda crystals which are left exposed to the air. Reason (R) : Washing soda is a hygroscopic substance.

(c) A is true but R is false. (Washing soda is efflorescent, not hygroscopic!) 🌬️

(iv) Assertion (A) : A crusty 'boiler scale' is formed in boilers when hard water is used. Reason (R) : Hard water contains bicarbonates of calcium and magnesium.

(a) Both A and R are true and R is the correct explanation of A. 🏭🪨

(v) Assertion (A) : Efflorescence is minimum during humid conditions. Reason (R) : Efflorescence occurs when the vapour pressure of the hydrated crystals exceeds the vapour pressure of the atmospheric humidity.

(a) Both A and R are true and R is the correct explanation of A. 🌧️💎

(vi) Assertion (A) : As the temperature is raised, the amount of solute that can be dissolved in a given quantity of solvent also increases. Reason (R) : A solution which is saturated at a given temperature, becomes unsaturated at a higher temperature.

(a) Both A and R are true and R is the correct explanation of A. 🌡️

(vii) Assertion (A) : Phosphorus when dissolved in carbon disulphide is a non aqueous solution. Reason (R) : Solutions of substances in a solvent other than water are called non aqueous solutions.

(a) Both A and R are true and R is the correct explanation of A. 🧵

(viii) Assertion (A) : Concentrated sulphuric acid absorbs moisture from the atmosphere. Reason (R) : Conc. H_2SO_4 is a deliquescent substance.

(c) A is true but R is false. (It is a hygroscopic liquid, not a solid that turns into a solution via deliquescence). 💧

(ix) Assertion (A) : Hardness of water is removed by boiling. Reason (R) : Water contains dissolved calcium and magnesium chlorides.

(d) A is false but R is true. (Only temporary hardness is removed by boiling; chlorides cause permanent hardness which boiling can't fix). ♨️🚫

(G) CASE STUDY BASED QUESTIONS

1. Imaad took a few blue crystals of copper sulphate. On heating the crystals, he observed that the blue colour disappeared and the substance turned into a white powder...

(a) How many water molecules are present in one formula unit of hydrated copper sulphate?

5 water molecules. 💧

(b) Name the white powder formed after heating copper sulphate crystals.

Anhydrous copper (II) sulphate. 🤍

(c) How can Imaad obtain the blue colour again from the white powder? Write the chemical equation for the reaction.

Simply by adding water to it!

Equation: $CuSO_4 + 5H_2O \rightarrow CuSO_4 \cdot 5H_2O$ 💙💦

(d) Write the chemical formula of gypsum. Name the product formed when gypsum is heated at 100°C.

Formula: $CaSO_4 \cdot 2H_2O$. When heated, it forms Plaster of Paris. 🩼

2. Srishti needs to dry ammonia gas in the laboratory. She knows the following drying agents...

(a) Why is solid calcium oxide preferred over other drying agents for drying ammonia gas?

Because ammonia is a basic gas, and calcium oxide is a basic drying agent. They will not react with each other. 🤝

(b) Why is concentrated sulphuric acid not used to dry ammonia gas?

Because concentrated sulphuric acid is highly acidic and will immediately react with the basic ammonia gas to form a completely different salt (ammonium sulphate). 💥 🛑

(c) Concentrated sulphuric acid is a good dehydrating agent. Explain.

It has a massive chemical affinity for water. It can literally strip hydrogen and oxygen atoms (in the 2:1 ratio of water) out of the chemical structure of other compounds! 🧲 💦

(d) State one difference between a drying agent and a dehydrating agent.

A drying agent (like CaO) only removes physical, external moisture without altering the chemical. A dehydrating agent (like conc. H_2SO_4) chemically tears water molecules directly out of a compound's formula! ✂️ 🧪

3. A solution of a solid in a liquid (such as water) can be prepared by adding the solid to water with constant stirring...

(a) The correct order of the solubilities of these substances in water is:

Looking at a standard temperature column like 286K: Y (30) > Z (24) > X (20). So the order is (iv) Y > Z > X. 📊

(b) 10 g of a solute is dissolved in 250 g of a solvent. The solubility of the solute is:

Solubility = (10g / 250g) * 100 = 4 g. Therefore, option (i) 4 g. ➗

(c) A saturated solution becomes unsaturated by:

(c) (i) Heating the solution AND (iv) Adding more of the solvent. 🔥 💧

(d) Solubility of a solute in a solvent may be defined as:

(iv) The maximum amount of solute which can be dissolved in 100 g of the solvent to form a saturated solution at a given temperature. 📏

(H) HOTS QUESTION

1. Water acts as a universal solvent because it has a unique property to dissolve various organic and inorganic compounds...

(a) What properties of water make it an exceptional solvent compared to other solvents like ethanol, carbon tetrachloride, benzene, etc.

Water has an exceptionally high dielectric constant and a highly polar covalent nature. This allows it to easily break the electrostatic forces binding ions in ionic compounds, effectively pulling them apart and dissolving them. ⚡🧲

(b) Explain the environmental consequences of water's "universal solvent" property, particularly regarding pollution.

Because water is so good at dissolving things, it rapidly dissolves toxic industrial chemicals, heavy metals, agricultural pesticides, and fertilizers. As it flows, it transports these hidden dissolved toxins straight into rivers, lakes, and groundwater, causing catastrophic widespread pollution and eutrophication that destroys ecosystems. ☠️🌍

Additional Application Questions (From Image 5)

2. A group of students took an old shoe box and covered it with a black paper from all sides. They fixed a source of light...

(a) Explain why the milk sample was illuminated. Name the phenomenon involved.

Milk is a colloid, meaning its suspended particles are large enough to scatter the beam of light, making its path glowing and visible. This specific phenomenon is called the Tyndall effect. 🔦🥛

(b) Same results were not observed with a salt solution. Why?

A salt solution is a true solution. Its particles are broken down to incredibly tiny sizes (less than 1 nm), which are way too small to scatter light rays. 🧂🚫

(c) Can you suggest two more solutions which would show the same effect as shown by the milk solution?

Starch solution and Soap solution (both are colloids!). 🥔 🧼

3. Name some physical and chemical processes used to make river water potable.

Physical processes: Sedimentation and Filtration.

Chemical processes: Coagulation (loading with alum) and Chlorination (adding chlorine to kill bacteria). 🏭 💧

4. Which steps are most effective at removing solid impurities from water? Explain the role of alum.

Filtration and Sedimentation are the most effective steps for removing solids. Alum acts as a coagulant. It attaches to extremely fine, suspended clay/mud particles, clumping them together to make them heavy so they sink rapidly to the bottom (a process called loading). 🪨 🔽

5. Which of the following is a primary indicator for sustainable development goal regarding drinking water?

(b) Percentage of population with access to safe and affordable drinking water. 🚰 🌍